

Soumya Das

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Summary

Edge AI developer and CS graduate student (CGPA **8.54/10**) with production experience deploying quantized CNN models on **NVIDIA Jetson Orin Nano** via CUDA, achieving a **4× GPU speedup** and sub-100 ms inference latency. Designs end-to-end IoT-ML pipelines — from wearable sensor integration to Django REST back-ends — with real-world deployment across agriculture and healthcare. Published oral presenter at **ICPAT 2025**; Team Leader in 4 national hackathons including **Smart India Hackathon 2025**. Targeting roles in **Machine Learning Engineering, Embedded AI, and IoT Systems**.

Technical Skills

ML / AI	Python, TensorFlow Lite, CNN, Model Quantization (FP16/INT8), Computer Vision, LangChain, Scikit-learn, Applied AI
Edge / Embedded	NVIDIA Jetson Orin Nano, CUDA, TensorRT-ready pipelines, Raspberry Pi, IoT sensor integration, hardware-software interfacing, Linux
Languages	Python (primary), C, C++, Java, JavaScript, PHP, SQL
Web / Backend	Django, REST APIs, Node.js, React.js, PHP/Core PHP, ASP.NET, AJAX, Bootstrap, Tailwind CSS
Databases	MySQL, MongoDB
Tools	Git, GitHub, VS Code, PyCharm, Figma, Draw.io, XAMPP, n8n
Concepts	OOP, Data Structures & Algorithms, System Design, Agile, Webhook Pipelines, Prompt Engineering

Experience

AI & IoT Innovation Intern | *Tinkering Hub, Parul University* | Vadodara, India Jan 2026 – Present

- Identified root-cause inference bottleneck in a live AI system; **rewrote ML inference pipeline**, slashing latency from **800 ms** → **<200 ms** — a **75% reduction** and ~40% overall efficiency gain.
- Deployed **FP16-quantized CNN models** (TensorFlow Lite) on NVIDIA Jetson Orin Nano via CUDA; benchmarked **4×+ GPU speedup** vs. CPU-only Raspberry Pi baseline, enabling real-time inference at the edge.
- Architected an **AI Crop & Soil Field Diagnosis System** using OOP Python, Django REST APIs, and IoT sensor fusion — achieving **96%+ disease detection accuracy** across **3+ live farming deployments**.
- Authored full system-architecture documentation, hardware integration specs, and deployment runbooks; collaborated with hardware and software stakeholders to ship production-grade AI-hardware solutions on schedule.

Data & Automation Intern | *HireZapp Technologies* | Remote Nov 2025 – Jan 2026

- Engineered **Python automation scripts** eliminating manual processing across **5+ recurring data pipelines**, cutting turnaround time significantly.
- Validated and cleansed large-scale datasets, maintaining accuracy and consistency across multiple product lines; **received Letter of Recommendation** for outstanding analytical contribution.

Projects

Edge-AI Wearable Smart Health Monitoring System Dec 2025 – Present

Python • CNN • TensorFlow Lite • CUDA • NVIDIA Jetson Orin Nano • IoT 🏠

- Problem:** Wearable patient-monitoring on CPU hardware produced unacceptable latency (>800 ms) for real-time clinical alerting.
- Deployed **FP16 + quantized INT8 CNN models** on Jetson Orin Nano via CUDA, achieving **97–98% classification accuracy** and **<100 ms end-to-end inference** — a **4×+ speedup** over Raspberry Pi CPU baseline.
- Integrated wearable IoT sensor array with a real-time ML inference pipeline, enabling **automated anomaly alerts** and continuous remote monitoring without cloud dependency.
- Independently managed full lifecycle: dataset curation, model training, CUDA optimisation, hardware flashing, benchmarking, and technical documentation.

AI-Powered Crop Health & Precision Agriculture System

Sep 2025 – Present

Python • CNN • Django • REST APIs • IoT Sensors • Machine Learning 

- **Problem:** Manual crop disease identification and fertiliser dosing caused ~35% avoidable seed waste and 25–30% fertiliser overuse.
- Trained a multi-class CNN on **1,500+ field images**, reaching **96%+ disease detection accuracy** and **~92% seed classification accuracy**.
- Built soil-analysis (pH, NPK), fertiliser-calculator, and weather-API modules that reduced fertiliser overuse by **25–30%** and improved crop-planning efficiency by **~40%**.
- Delivered a Django-based farmer portal with an agri-marketplace listing **50+ products**, cutting manual decision-making time by **~60%** — live across **3+ farming deployments**.

Rural Banking Platform with AI Fraud Detection

Sep 2024 – Present

PHP • MySQL • JavaScript • AJAX • Supervised & Unsupervised ML

- Built a secure internet-banking system with **ML-based fraud detection** combining supervised classification and unsupervised anomaly detection, multi-level authentication, and real-time fund-transfer modules.

Learning Management System — Rural Education Platform

Jun – Aug 2025

PHP • MySQL • JavaScript • AJAX • Bootstrap 

- Engineered a full-stack LMS serving **200+ rural students** with AJAX async rendering (**40% UI performance gain**), role-based auth, attendance tracking, file uploads, and live-session support.

Research & Leadership

- **Oral Presenter & Published Author — ICPAT 2025** (International Conference on Power & Advanced Technologies): Presented paper “*Smart Automotive Vehicles: Intelligent Systems & Real-Time Decision Making*” to **100+ international delegates**; paper accepted and published in conference proceedings.
- **Team Leader — 4 National Hackathons:** Smart India Hackathon 2025, PU Code Hackathon 3.0, Tech Expo, Product Management Training — led **4–6 member cross-functional teams** from ideation through system design to final live demo.
- **Certificate of Highest Attendance** — Parul University; active on **LeetCode** (competitive programming).

Education

Integrated MCA (B.C.A. + M.C.A., equivalent to Master’s in CS) | *Parul University, Vado-* Jul 2024 – Jul 2028
dara | CGPA **8.54/10**

AICTE/UGC Recognised | Coursework: DSA, OOP, Database Systems, Software Engineering, System Design, Computer Networks, Applied AI

Higher Secondary Certificate (Computer Science) | *Saharda Kalipada Vidyapith, West Ben-* Mar 2022 – Jun 2024
gal | Score **69.5%**

Certifications

- **Oracle Fusion AI Agent Studio** — Foundations Associate
- **IBM Python for Data Science**
- **NPTEL Programming in Java** — *Elite* Grade
- **Cisco Python Essentials 1 & 2**
- **Microsoft Foundational C#**
- **Google Analytics Certification**
- **Certiport ITS JavaScript**
- **Certiport ITS HTML & CSS** | **IBM SQL 101**